STA2002: Probability and Statistics II
Confidence Interval-Part II

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Introduction

- In this lecture we will continue our discussion on confidence intervals.
- Our focus today is on confidence intervals for difference of two means, confidence intervals for proportions, and sample size required for confidence intervals within a given margin of error.
- **Suggested reading**: Chapter 7.2, 7.3 and 7.4 of the textbook.

C.I. for the Difference of Two Means

- Suppose that we have two populations from which we draw i.i.d. random samples, say $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$. We would like to develop confidence intervals for $\mu_X - \mu_Y$, the difference between the two population means.
- Consider the following cases:
  1. Case 1: $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ are independent
     (a) $\sigma_X^2 = \sigma_Y^2 = \sigma^2$, and σ^2 is unknown (Two populations have the same variance σ^2)
     (b) $\sigma_X^2 \neq \sigma_Y^2$, and they are unknown
  2. Case 2: $n = m$ and X_i, Y_i are dependent for $i = 1, \ldots, n$. However, the n pairs $(X_1, Y_1), \ldots, (X_n, Y_n)$ are independent.

Case 1(b): Welch's t-interval

Theorem 1

Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu_X, \sigma_X^2)$ *and* $Y_1, \ldots, Y_m \overset{i.i.d.}{\sim} N(\mu_Y, \sigma_Y^2)$ *be independent random variables. Then an approximate* $100(1-\alpha)\%$ *C.I. for* $\mu_X - \mu_Y$ *is*

$$\overline{X} - \overline{Y} \pm t_{\alpha/2}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}},$$

where

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor.$$

Remark. For any real number x, $\lfloor x \rfloor$ is the floor function of x, which is the greatest integer less than or equal to x. For example, $\lfloor 3.4 \rfloor = \lfloor 3.9 \rfloor = \lfloor 3 \rfloor = 3$.

Example

Example 2

Recall Example 7.2-2 in the textbook. This time we assume $\sigma_X^2 \neq \sigma_Y^2$. Let $X \sim N(\mu_X, \sigma_X^2)$ be the score on a standardized test in a large high school, and $Y \sim N(\mu_Y, \sigma_Y^2)$ be the score on a standardized test in a small high school. We have

$$
\begin{aligned}
n &= 9 \\
\overline{x} &= 81.31 \\
s_x^2 &= 60.76 \\
m &= 15 \\
\overline{y} &= 78.61 \\
s_y^2 &= 48.24.
\end{aligned}
$$

Example

Solution. To construct a 95% Welch's t-interval for $\mu_X - \mu_Y$, we calculate

$$r = \left\lfloor \frac{\left(\frac{s_x^2}{n} + \frac{s_y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{s_x^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{s_y^2}{m}\right)^2} \right\rfloor = 15$$

$$t_{0.025}(r) = 2.131$$

$$\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}} = \sqrt{\frac{60.76}{9} + \frac{78.61}{15}} = 3.4629$$

The required 95% Welch's t-interval interval for $\mu_X - \mu_Y$ is

$$81.31 - 78.61 \pm 2.131 \times 3.4629 = [-4.6794, 10.0794].$$

This should be compared with the two-sample pooled t-interval calculated in Lecture Slides 4:

$$[-3.65, 9.05]$$

□

Case 2: Paired t-interval

Theorem 3

Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu_X, \sigma_X^2)$ *and* $Y_1, \ldots, Y_n \overset{i.i.d.}{\sim} N(\mu_Y, \sigma_Y^2)$. *Let*

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$$

be n independent pairs, but X_i *and* Y_i *for a given i are dependent. Also let* $D_i = X_i - Y_i$ *for* $i = 1, \ldots, n$. *Then a* $100(1-\alpha)\%$ *C.I. for* $\mu_X - \mu_Y$ *is*

$$\overline{D} \pm t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}},$$

where

$$\overline{D} = \frac{1}{n}\sum_{i=1}^{n} D_i$$

$$S_D^2 = \frac{1}{n-1}\sum_{i=1}^{n}(D_i - \overline{D})^2.$$

Case 2: Paired t-interval

Proof. We know that $D_1, \ldots, D_n$ are independent normal random variables with mean $\mu_X - \mu_Y$. As a result,

$$\frac{\overline{D} - (\mu_X - \mu_Y)}{S_D/\sqrt{n}} \sim t(n-1).$$

As a result,

$$P\left(\overline{D} - t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}} \leq \mu_X - \mu_Y \leq \overline{D} + t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{b}}\right) = 1 - \alpha$$

□

Case 2: Paired t-interval

Example 4

This is Example 7.2-4 in book. Let X_i be the reaction time to red light, and Y_i be the reaction time to green light. From the data presented in the book, we have

$$\begin{aligned} n &= 8, \\ t_{0.025}(7) &= 2.365, \\ \overline{d} &= -0.0625, \\ s_d &= 0.0765. \end{aligned}$$

The 95% C.I. for $\mu_X - \mu_Y$ is

$$-0.0625 \pm 2.365 \frac{0.0765}{\sqrt{8}} = [-0.1265, 0.0015].$$

C.I. for Proportions

- Let p be the proportion of population who has a certain characteristic.
- To estimate p, we draw n i.i.d. random samples $X_1, \ldots, X_n \overset{i.i.d.}{\sim} \text{Bernoulli}(p)$, where each X_i is 1 if the individual has the characteristic, and is 0 if the individual does not have that characteristic.
- Denote $Y = \sum_{i=1}^{n} X_i$, then $E(Y) = np$ and $\text{Var}(Y) = np(1-p)$.
- Moreover, if n is large enough, then by central limit theorem, we have

$$\frac{Y}{n} = \frac{\sum_{i=1}^{n} X_i}{n} \overset{\text{approx}}{\sim} N\left(p, \frac{p(1-p)}{n}\right).$$

C.I. for Proportions

- Recall that the mle of p is,

$$\hat{p} = \frac{Y}{n} = \frac{\sum_{i=1}^{n} X_i}{n}.$$

- The resulting approximate $100(1-\alpha)\%$ C.I. for p is

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

C.I. for Proportions

- Note that, to construct an approximate $100(1-\alpha)\%$ C.I. for p, for large enough n, we have that,

$$P\left(-z_{\alpha/2} \leq \frac{\hat{p}-p}{\sqrt{\frac{p(1-p)}{n}}} \leq z_{\alpha/2}\right) \approx 1-\alpha$$

$$P\left(\hat{p}-z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \leq p \leq \hat{p}+z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right) \approx 1-\alpha.$$

- As p is unknown, when n is large enough, $\hat{p}$ is used to approximate p.

C.I. for Proportions

Example 5

This is Example 7.3-2 in book. In a survey of a political campaign, there are $y = 185$ out of $n = 351$ voters who favor a candidate. We then calculate $\hat{p} = 185/351 = 0.527$, and an approximate 95% C.I. for p is

$$0.527 \pm 1.96\sqrt{\frac{0.527(1-0.527)}{351}} = [0.475, 0.579].$$

C.I. for Proportions

- Suppose now that we have two populations instead of one.
- Let p_i be the proportion of population i who has a certain characteristic, and denote by Y_i to be the number of successes in population i for $i = 1, 2$.
- Let n_i to be the number of samples in population i.
- Since $E(Y_i/n_i) = p_i$ and $\mathrm{Var}(Y_i/n_i) = p_i(1-p_i)/n_i$ and they are independent, we have

$$E\left(Y_1/n_1 - Y_2/n_2\right) = p_1 - p_2,$$

$$\mathrm{Var}\left(Y_1/n_1 - Y_2/n_2\right) = \frac{p_1\left(1-p_1\right)}{n_1} + \frac{p_2\left(1-p_2\right)}{n_2}.$$

C.I. for Proportions

- To construct an approximate $100(1-\alpha)\%$ C.I. for $p_1 - p_2$, for large enough n_1 and n_2 we have that,

$$P\left(-z_{\alpha/2} \leq \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \leq z_{\alpha/2}\right) \approx 1 - \alpha$$

- Again, we further approximate p_i by $\hat{p}_i$. The resulting approximate $100(1-\alpha)\%$ C.I. for $p_1 - p_2$ is

$$\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}.$$

C.I. for Proportions

Example 6

Let p_1 be the proportion of male students into gaming, and p_2 be the proportion of female students into gaming. Suppose that $\hat{p}_1 = 0.6, \hat{p}_2 = 0.4$. If $n_1 = 2000, n_2 = 2200$, then an approximate 95% C.I. for $p_1 - p_2$ is

$$0.6 - 0.4 \pm 1.96 \sqrt{\frac{0.6(1-0.6)}{2000} + \frac{0.4(1-0.4)}{2200}} = [0.17, 0.23].$$

Sample Size Determination

- Suppose that $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$, and σ^2 is known.
- The $100(1-\alpha)\%$ C.I. for μ is $\overline{x} \pm z_{\alpha/2}\sigma/\sqrt{n}$.
- For any given $\epsilon > 0$, in practice we would like to determine the sample size required so that our C.I. is

$$\overline{x} \pm \epsilon.$$

- As a result, we set,

$$\epsilon = \frac{z_{\alpha/2}\sigma}{\sqrt{n}},$$

$$n = \frac{z_{\alpha/2}^2\sigma^2}{\epsilon^2}.$$

Sample Size Determination

- One half the length of the C.I. is called the **maximum error of the point estimate**.
- If σ^2 is unknown, then the $100(1-\alpha)\%$ C.I. for μ is $\overline{x} \pm t_{\alpha/2}(n-1)s/\sqrt{n}$, where s^2 is the sample variance.
- We set

$$\epsilon = \frac{t_{\alpha/2}(n-1)s}{\sqrt{n}},$$

$$n = \frac{t_{\alpha/2}(n-1)^2 s^2}{\epsilon^2}.$$

- However, this expression has two problems:
  - $t_{\alpha/2}(n-1)$ depends on n. We shall use $z_{\alpha/2}$ instead.
  - s^2 depends on n. We can estimate s^2 by existing or expert knowledge.

Sample Size Determination

- For proportion, we can also carry out a similar calculation.
- An approximate $100(1-\alpha)\%$ C.I. for p is

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

We set

$$\epsilon = z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}},$$

$$n = \frac{z_{\alpha/2}^2 \hat{p}(1-\hat{p})}{\epsilon^2}.$$

Sample Size Determination

- Note that in the expression, $\hat{p}$ depends on n. There are two possible solutions:

  (a) We know that the true value of p is around a value, say p^*, then we can use

  $$n = \frac{z^2_{\alpha/2} p^*(1-p^*)}{\epsilon^2}.$$

  (b) For any $p \in [0,1]$, we can show that $p(1-p) \leq \frac{1}{4}$. Making use of this, the required sample size is

  $$n = \frac{z^2_{\alpha/2}}{4\epsilon^2}.$$